1. Define kinetic energy.
2. Define potential energy.
3. What is the formula for kinetic energy?
4. What is the formula for potential energy?
5. Determine the amount of potential energy a 1kg book has on the shelf of a bookcase that is
    a. 1.0m high
    b. 1.5m high
    c. 2.0m high
6. You are on roller skates at the top of a hill. Your potential energy is 1000J and your mass is 60kg.
    a. What is the height of the hill?
    b. If you roll down the hill, your PE is converted to KE. At the bottom of the hill, your KE will be equal to your PE at the top. Calculate your speed at the bottom of the hill.
7. A 1.0kg ball is thrown in the air with a velocity of 30m/s. How much kinetic energy does it have?
8. What is the kinetic energy of a boat moving t 5m/s if the boat's mass is 2000kg?
9. What is the mass of an object travelling at 30m/s if it has 33,750J of energy?
10. A 2 kg ball rolling along a flat surface starts up a hill. If the ball reaches a height of 0.63 m, what was its initial speed?